

SmartGym Adaptive Training System

Readme

SmartGym project documentation.

Version 1.0 Prototype

ESP32-S3 firmware | Firebase RTDB | React dashboard

Generated documentation package

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SmartGym Documentation

This folder contains the documentation package for the SmartGym v1.0 prototype. Each major guide is available as editable Markdown and as a formatted PDF.

Start Here

- [Project README](../README.md)
- [User Manual](USER_MANUAL.md) | [PDF](USER_MANUAL.pdf)
- [System Architecture](SYSTEM_ARCHITECTURE.md) | [PDF](SYSTEM_ARCHITECTURE.pdf)
- [Developer Guide](DEVELOPER_GUIDE.md) | [PDF](DEVELOPER_GUIDE.pdf)
- [Firebase Guide](FIREBASE_GUIDE.md) | [PDF](FIREBASE_GUIDE.pdf)
- [Troubleshooting Guide](TROUBLESHOOTING.md) | [PDF](TROUBLESHOOTING.pdf)

Reference Documents

| Document | Markdown | PDF |

| --- | --- | --- |

| Firebase Webapp Schema | [Open](FIREBASE_WEBAPP_SCHEMA.md) |
[PDF](FIREBASE_WEBAPP_SCHEMA.pdf) |

| Firebase Dashboard Seed Guide | [Open](FIREBASE_DASHBOARD_SEED_GUIDE.md) |
[PDF](FIREBASE_DASHBOARD_SEED_GUIDE.pdf) |

| Migration Notes | [Open](MIGRATION_NOTES.md) | [PDF](MIGRATION_NOTES.pdf) |

| SquareLine Handoff | [Open](SQUARELINE_HANDOFF.md) |
[PDF](SQUARELINE_HANDOFF.pdf) |

Recommended Reading Order

1. Read the [User Manual](USER_MANUAL.md) to understand how the device is used.
2. Read the [System Architecture](SYSTEM_ARCHITECTURE.md) for the full data flow.
3. Read the [Developer Guide](DEVELOPER_GUIDE.md) before changing firmware.
4. Read the [Firebase Guide](FIREBASE_GUIDE.md) before importing, editing, or migrating database data.
5. Use the [Troubleshooting Guide](TROUBLESHOOTING.md) during demos and hardware testing.

Regenerating PDFs

Run this from the repository root:

```
python tools/generate_docs_pdfs.py
```